6 Week Python + GPIO + Electronics

Raspberry Pi 5 • T Cobbler • Breadboard • gpiozero

Hands on, project based course for S1-S3

Includes weekly worksheets, code, wiring diagrams, and extension tasks

Prepared for classroom use: printable and student friendly.

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General Wiring Reference (Pi 5 → T Cobbler → Breadboard)

```
RASPBERRY PI 5 – 40 pin GPIO Header (Top View)
-----------------------------------------------
 (1)3V3   (2)5V
 (3)GPIO2 (4)5V
 (5)GPIO3 (6)GND
 (7)GPIO4 (8)GPIO14
 (9)GND   (10)GPIO15
 (11)GPIO17 (12)GPIO18
 (13)GPIO27 (14)GND
 (15)GPIO22 (16)GPIO23
 (17)3V3    (18)GPIO24
 (19)GPIO10 (20)GND
 (21)GPIO9  (22)GPIO25
 (23)GPIO11 (24)GPIO8
 (25)GND    (26)GPIO7
 (27)ID_SD  (28)ID_SC
 (29)GPIO5  (30)GND
 (31)GPIO6  (32)GPIO12
 (33)GPIO13 (34)GND
 (35)GPIO19 (36)GPIO16
 (37)GPIO26 (38)GPIO20
 (39)GND    (40)GPIO21
-----------------------------------------------
T COBBLER maps these pins to breadboard rows. Use 3V3 and GND rails; never feed 5V into a GPIO pin.
```

Safety notes: Use 3.3V logic only. Always include appropriate resistors for LEDs and sensor voltage dividers.

Hardware used

RasPi5 - 8Gb kit

T-Cobbler Assembled

LCD Waveshare 1602

Ultrasonic distance sensor

Potentiometer 10k

MCP3008 Anal-Digi converter

Breadboard

Buzzer

Buttons

LED 5mm

Jumper Wires

Resistors 1/2 Watt - 220 ohm

Project 1 — Intro to Python & Raspberry Pi

Learning Objectives

- Run simple Python programs in Thonny
- Use print(), input(), and variables
- Identify GPIO pins on the Pi 5 via the T Cobbler

Wiring Diagram (ASCII)

Orientation only this this time — no live circuit.
Map rails: 3V3 → red rail, GND → blue rail. Label GPIO26 (LED), GPIO21 (button) for later.

Example Code

Hello World

```
print("Hello, world!")
```

Variables

```
name = "Thomas"
print("My name is", name)
```

User Input

```
name = input("What is your name? ")
print("Nice to meet you,", name)
```

Mini Project — Name Badge

```
name = input("Enter your name: ")
print("**********************")
print("*     " + name + "     *")
print("**********************")
```

Extension Tasks

- Typing speed tester
- ASCII animation
- Random maths quiz
- Menu to choose different badge borders

More Examples

```
my_number = 7
my_string = "hello"
print(my_string,my_number)

my_age = input("What is your age? ")
print(my_age*3)
print(int(my_age*3))

my_age_num =int(input("What is your age? "))
print(my_age_num*3)

total = 0
total = total + 1
total = total + 1
total = total + 1
print(total)

sunny = False
if sunny:
    print("Let's go to the park")
else:
    print("We must stay at home")

print("How old are you?")
age = int(input())
print("How tall are you?")
height = int(input())
if age > 12:
    print("You can ride")
elif height > 150:
    print("You can ride")
else:
    print("YOU MAY NOT RIDE, GO AWAY!")

for x in range(0, 11):
    print(x)

for a in range(0, 6):
    for b in range(0, 6):
                print(a, "times", b, "is", a * b)

def my_func():
    print("This is my function")
    print("Imagine there was lots of code here")
    print("that you didn't want to type many times")
my_func()
my_func()
my_func()
```

Project 2 — LEDs & Outputs

Learning Objectives

- Understand LED polarity and resistors
- Control an LED via GPIO output
- Use loops and timing

Project 2.1

Blink LED

As in the pictures run the blue wire from the GND port to any contact point on the blue marked column.

Now add a resistor from a contact point in the blue column to the first point on a row. Take the Red LED and put the **short leg** on the same row as the resistor. The long leg should go on the row below. Now take another cable and run it from the same row as the long leg and take it to #13 on the cobbler

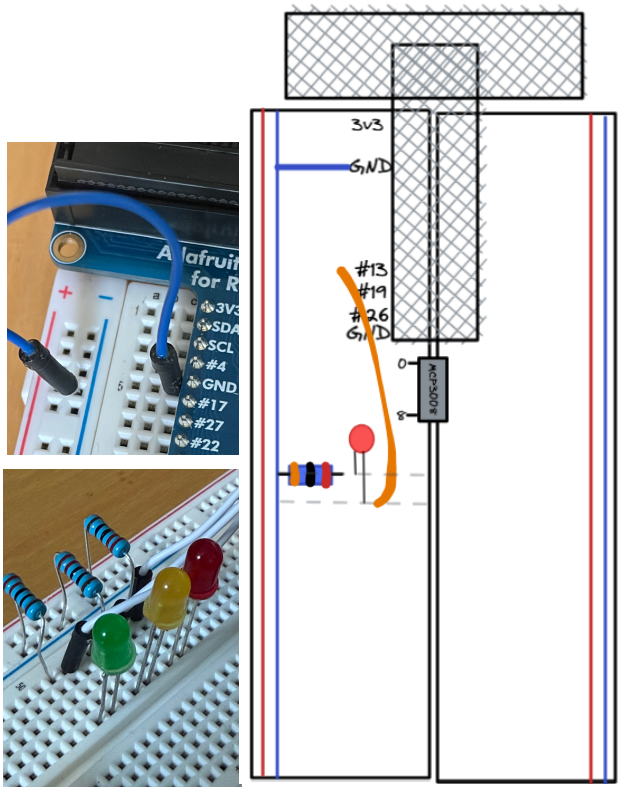
Write the following code in Thonny

```
from gpiozero import LED

from time import sleep

led = LED(13)

while True:
      led.on()
      sleep(0.5)
      led.off()
      sleep(0.5)
```



Project 2.2 — Traffic Light

Add an AMBER and GREEN light to the breadboard similar to the RED light
Use #19 for Amber and #26 for green
Write and run the following code in Thonny but add the code for the sequence

```
from gpiozero import LED
from time import sleep
# Define the LEDs connected to GPIO pins
red = LED(13)
amber = LED(19)
green = LED(26)
while True:
      # Red light on for 5 seconds
      red.on()
      amber.off()
      green.off()
      sleep(5)
      # Red and amber lights on for 2 seconds

      # Amber light on for 1 second

      # Green light on for 6 seconds

      # Amber light on for 1 second
```

Project 3 — Buttons & Digital Inputs

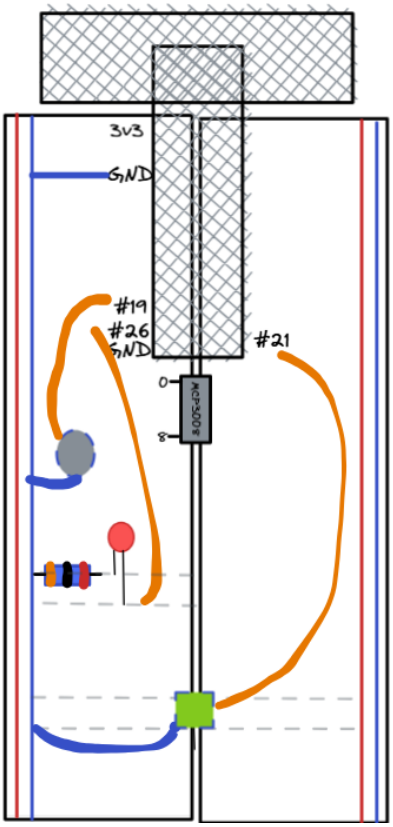
Learning Objectives

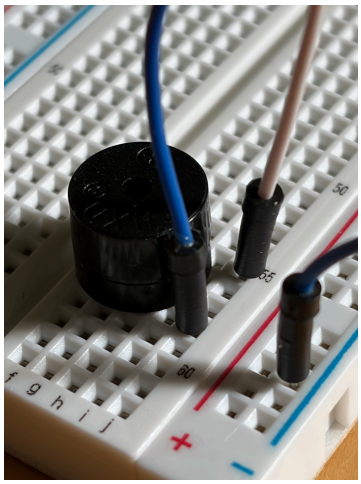
- Read digital inputs
- Use if statements to control outputs

Project 3.1 - Button

Button controls LED

As before - run the blue wire from GND to a contact point on the blue column. Add a LED with resistor. Place the button across the space on the breadboard. Push down to make sure it goes in. Take the other blue wire and run it from the same row as the bottom leg of the switch to any socket on the GND column. Run an orange cable from the top leg row to #26 on the cobbler. Write and run the code below in Thonny.





```
from gpiozero import LED, Button

led = LED(26)
button = Button(21)

while True:
     if button.is_pressed:
         led.on()
     else:
        led.off()
```

Project 3.2 — Buzzer

The buzzer is added the same as a switch but the short leg row links to GND. Run the orange cable from the long leg row to #5.
Write and run the following code in Thonny

```
from gpiozero import Buzzer
from time import sleep
bz = Buzzer(19)
bz.off()
bz.on()
sleep(0.2)
bz.off()
```

Project 3.3 — Buzzer and LED and Button

```
from gpiozero import LED, Button, Buzzer
from time import sleep, time
from signal import pause
bz = Buzzer(19)
bz.off()
led = LED(26)
led.off()
button = Button(21)
def buznlighton():
    led.on()
    bz.on()
def buznlightoff():
    led.off()
    bz.off()
button.when_pressed = buznlighton
button.when_released = buznlightoff
pause()
```

Notice in the above example we used a Function to have two things happen when the function is used. A function is created using a defentition in this format def <function>: and the commands under it are indented.

Project 3.4 — Send this Morse Code message

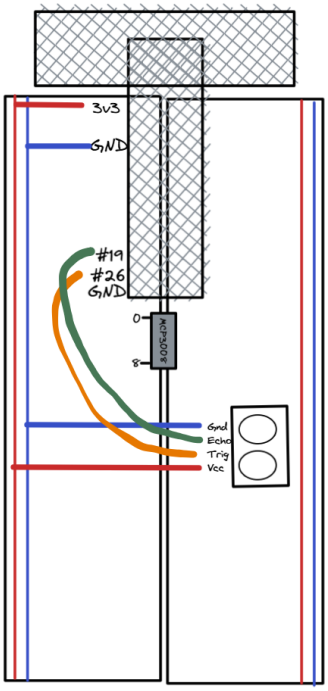
```
I      L      O      V      E      P      Y      T    H      O      N

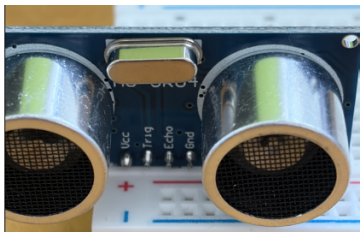
..     .-..   ---    ...-   .     .--.   -.--   -    ....   ---    -.
```

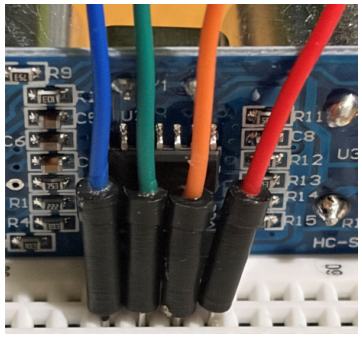
Project 4 — Sensors

Learning Objectives

- Understand analog vs digital sensors
- Read sensor values in Python
- Use thresholds to make decisions







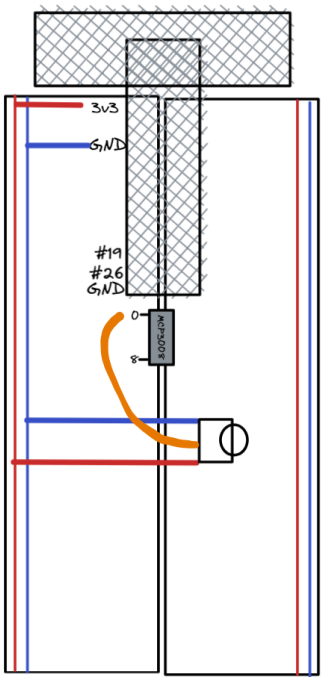
Project 4.1 - Ultrasonic Distance Sensor

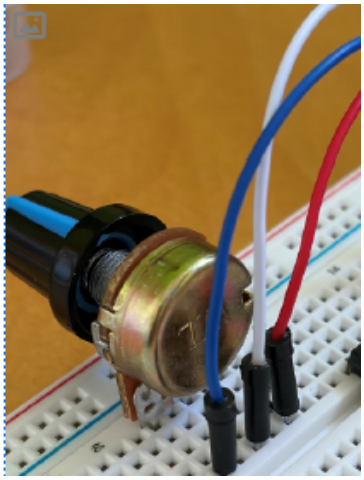
Wire up the sensor as the top set of pictures. This sensor sends out a ping from one of the elements Trig and times the return to the Echo- then calculates the distance to send as a digital value to the RasPi.

```
from gpiozero import DistanceSensor
from time import sleep

sensor = DistanceSensor(trigger=26, echo=19)

while True:
    print(sensor.distance * 100, "cm")
    sleep(0.5)
```





Project 4.2 — Potentiometer

This is a variable resistor and needs a chip to convert the analogue signals into values to send. Note the Orange cable plugs into a row with the chip not the cobbler.

```
from gpiozero import MCP3008
from time import sleep
pot = MCP3008(channel=0)
while True:
    level = pot.value
    if level < 0.3:
        print("light off:", round(level, 2))
    else:
        print("Light on:", round(level, 2))
    sleep(1)
```

Project 4.3

- **Add LED to trigger on distance or Potential**
- **Show output as a percentage**

Project 5 — Combining Inputs & Outputs

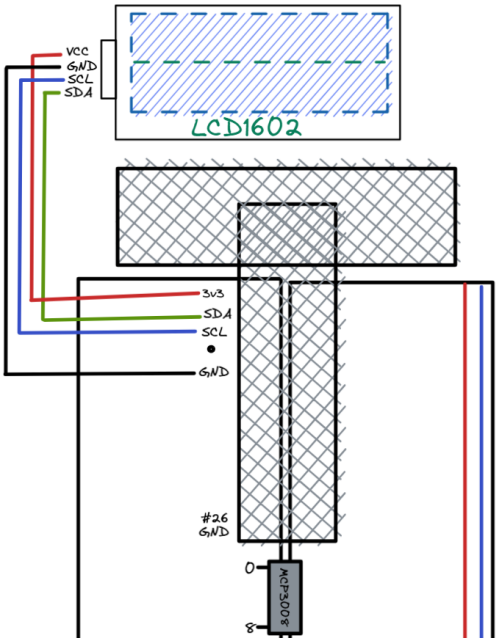
Learning Objectives

- What is the I2C interface
- Output text to the LCD
- Learn what other functions there are

Important !

Wiring Diagram

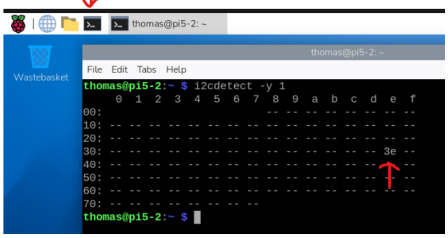
Connect The LCD while the RasPi is switched off.
The I2C interface detects devices on startup



Project 5.1

I2C (Inter-Integrated Circuit) is a synchronous, serial communication bus that allows multiple peripheral devices to communicate with a controller using just two wires: SDA (Serial Data) and SCL (Serial Clock). It is commonly used for short-distance communication within electronic devices I2C can host up to 128 devices with enough power - the Raspberry Pi has enough power to run 3 or 4 devices on its i2c interface. Open a Terminal using the shortcut on the OS Menu Bar and run the command below to see what the LCD1602 has been assigned to.

```
i2cdetect -y 1
```



Project 5.2 - Add text to the LCD screen*write this code then run it in Thonny*

```
import LCD1602
lcd=LCD1602.LCD1602(16,2)
backlight = LCD1602.SN3193()
backlight.set_brightness(50)
while True:
      # set the cursor to column 0, line 0
    lcd.setCursor(0, 0)
      #Add a line of text to the top
    lcd.printout("St Mungo's Academy")
      #set the cursor to column 0, line 0
    lcd.setCursor(0, 1)
       #Add a line of text to the bottom
    lcd.printout("Coding Club")
```

Project 5.3

Stop the code and then Run this in the Python (bottom)section of Thonny to list all the functions the module offers.

```
>>import LCD1602
>>help(LCD1602)
```

Scroll up through the list and see there is a clear and a ScrollDisplayLeft. We will add these to the code.

Project 5.4

Add an import of time for time.sleep later.
Update the code with a try/except so we can quit out of the loop cleanly using CTRL+C to run a clear
Add the scrollDisplayLeft into the loop and add more text to display

```
import LCD1602
import time
lcd = LCD1602.LCD1602(16,2)
backlight = LCD1602.SN3193()
backlight.set_brightness(50)

try:
    while True:
        lcd.scrollDisplayLeft()
        # set the cursor to column 0, line 1
        lcd.setCursor(0, 0)
        lcd.printout("St Mungo's Academy Code Club")

        lcd.setCursor(0, 1)

        lcd.printout("SMACC! All you need for fun!")
        time.sleep(0.4)

except(KeyboardInterrupt):
    lcd.clear()
    del lcd
```

Project 5.5

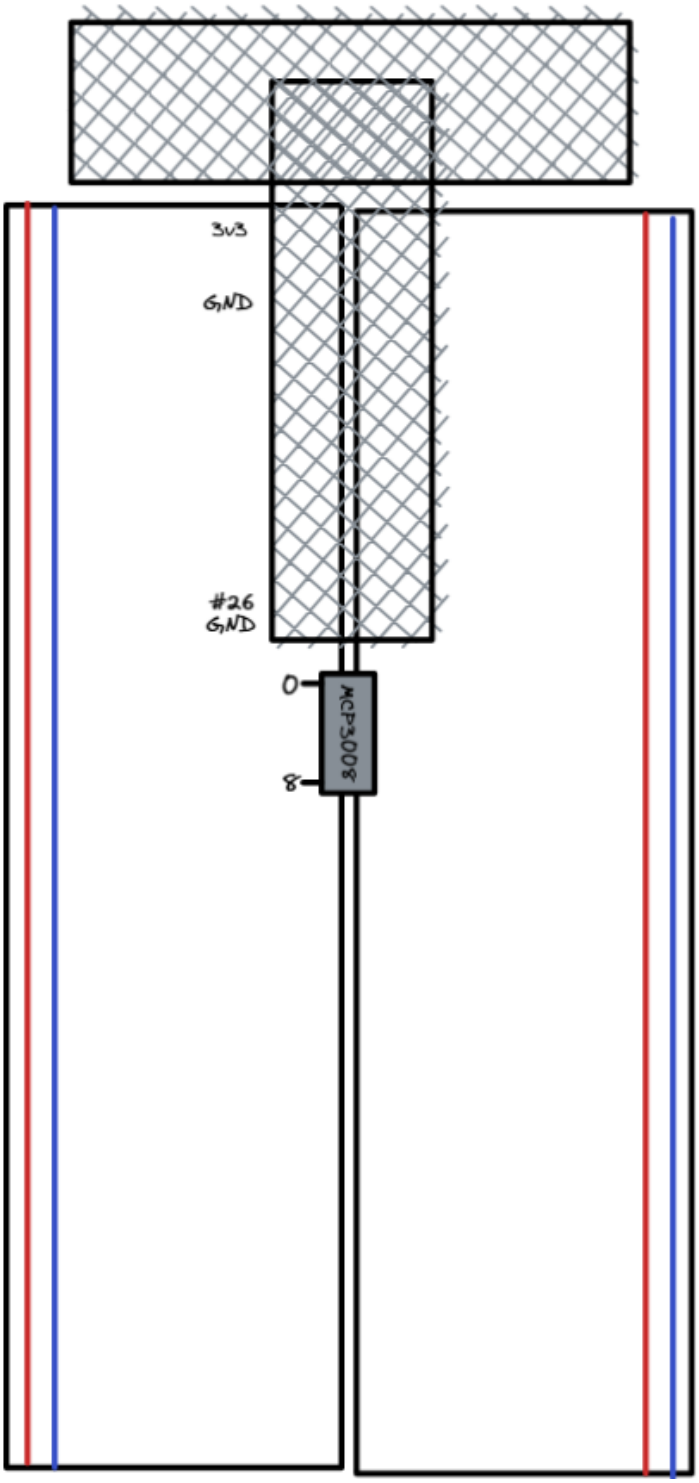
Use the LCD1602 with one of the other projects and put the output to the LCD screen.

Project 6 — Final Projects

Learning Objectives

- Plan and build an independent project
- Explain and demo the build
- Document code and wiring

Wiring Diagram



What will your project do?

What electronics will it use?

Python Code

Suggestions

- Buzzer with rate inverse proportional to distance. Parking sensor.
- Show potentiometer value using 6+ lights. Visual volume control.

Project — Showcase & Reflection

What went well?

What problems did you overcome?

What would you do differently next time?

Extension Tasks - how can your project be improved?

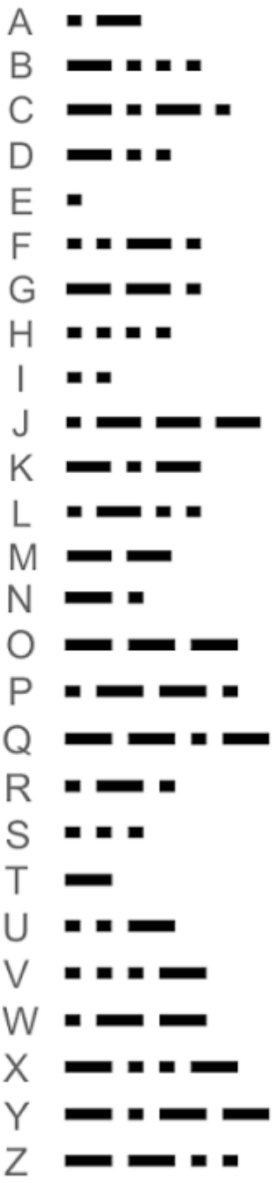
Appendix: Troubleshooting & Fun Stuff

Troubleshooting Checklist

- Verify GPIO pin numbers match code
- Confirm GND rail is connected to Pi GND
- LED has a 220Ω resistor in series
- Use 3.3V HC SR04
- Check for loose jumper wires and short circuits

Fun stuff

Morse Code Reference



Pygamezero
Pongy Game

Python Code

```
Import pgzero
WIDTH = 800
HEIGHT = 600

ball = Actor("ball", (400, 300))
ball.vx = 4
ball.vy = 4

paddle = Actor("paddle", (50, 300))
def draw():
    screen.fill("black")
    ball.draw()
    paddle.draw()

def update():
    ball.x += ball.vx
    ball.y += ball.vy

    if ball.y <= 0 or ball.y >= HEIGHT:
        ball.vy *= -1
    if ball.x <= 0 or ball.x >= WIDTH:
        ball.x = 400
        ball.y = 300

def on_mouse_move(pos):
    paddle.y = pos[1]
```

Pygamezero
Flappy Birdy

Python Code

```
Import pgzero
WIDTH = 400
HEIGHT = 600

bird = Actor("bird", (100, 300))
bird.vy = 0
gravity = 0.5
gap = 150
pipe_x = WIDTH
pipe_height = 150
def draw():
    screen.fill("lightblue")
    bird.draw()
    screen.draw.line((pipe_x, 0),
(pipe_x, pipe_height), "green")
    screen.draw.line((pipe_x,
pipe_height + gap), (pipe_x,
HEIGHT), "green")

def update():
    global pipe_x
    bird.vy += gravity
    bird.y += bird.vy
    pipe_x -= 3

    if pipe_x < 0:
        pipe_x = WIDTH

def on_mouse_down():
    bird.vy = -10
```